Soil Chemistry: pH Distribution

McKinley Martin

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This R Markdown file details creating a distribution plot using ggplot2 from a csv file. In particular, this script focuses on the environmental pH distribution of various soil samples. For more information on basic density charts with ggplot2, reference the following link: https://r-graph-gallery.com/21-distribution-plot-using-ggplot2.html

Start by loading in the ggplot2 package.

```r
library(ggplot2)
```

Next, read in the csv file as the function. The head() function allows one to preview the dataset without loading in the entire file.

```r
density_function <- read.csv("CIB_Soil_Chemistry_pH.csv")
head(density_function)
```

```
##        Sample  pH
## 1 DVS-2406106 6.8
## 2 DVS-2406109 6.6
## 3 DVS-2406112 6.7
## 4 DVS-2406115 6.0
## 5 DVS-2406118 6.4
## 6 DVS-2406119 6.2
```

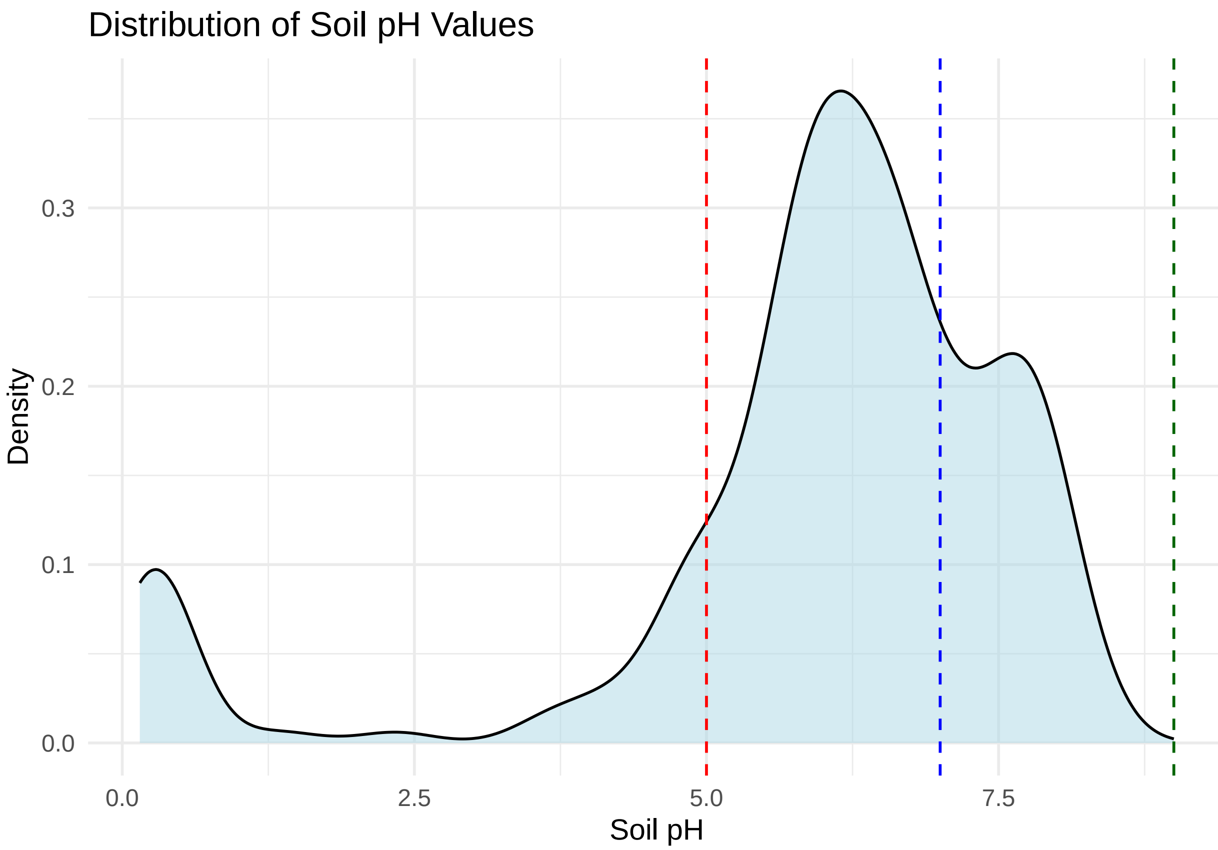
Finally, set up the script for running the plot. In this case, vertical lines were introduced to demonstrate experimental media pH values for evaluation against initial environmental conditions.

```r
ggplot(density_function, aes(x = pH)) +
  geom_density(fill = "lightblue", alpha = 0.5) +

  # create vertical lines
  geom_vline(xintercept = 5, linetype = "dashed", color = "red") +
  geom_vline(xintercept = 7, linetype = "dashed", color = "blue") +
  geom_vline(xintercept = 9, linetype = "dashed", color = "darkgreen") +

  theme_minimal() +

  # create labels
  labs(title = "Distribution of Soil pH Values", x = "Soil pH", y = "Density"
  )
```


Distribution of Soil pH Values
Density
0.3
0.2
0.1
0.0
0.0
2.5
5.0
7.5
Soil pH